Pandemic Disruptions and Post-COVID Patterns of Antimicrobial Resistance-Associated Infections in Sweden

Regional Incidence Patterns for ESBL/CARBA, VRE, and MRSA (2015-2025)

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1 Introduction

This analysis examines the impact of the COVID-19 pandemic on the epidemiology of three major antimicrobial resistance (AMR)-associated infections in Sweden: extended-spectrum beta-lactamase and carbapenemase-producing bacteria (ESBL/CARBA), vancomycin-resistant enterococci (VRE), and methicillin-resistant Staphylococcus aureus (MRSA). Using regional surveillance data from the Swedish Public Health Agency (Folkhälsomyndigheten), we compare incidence patterns across three distinct time periods:

- **Pre-COVID (2015-2019):** Baseline period before pandemic-related disruptions
- **COVID (2020-2021):** Period of pandemic restrictions including travel limitations and behavioral changes
- **Post-COVID (2022-2025):** Recovery period following relaxation of restrictions

We employ generalized linear models to estimate incidence rate ratios (IRRs) comparing these periods, providing insight into how pandemic-related disruptions affected the transmission dynamics of these AMR-associated infections.

Throughout, we frame the pandemic period as a **system-wide disruption** (affecting transmission, care-seeking, testing, and reporting) rather than a single causal exposure, and interpret changes as reflecting the combined influence of these mechanisms.

2 Methods

2.1 Data Sources

Annual regional case counts and incidence rates (per 100,000 population) for ESBL/CARBA, VRE, and MRSA were obtained from the Swedish Public Health Agency's national notifiable disease surveillance system (SmiNet) for the period 2015-2025 across all 21 Swedish counties. Population denominators were retrieved from Statistics Sweden (SCB) via the PxWeb API. For 2015-2024, annual county populations were obtained from SCB table `BefolkningNy` (contents `Folkmängd`) using **total age** and aggregated across sex and civil status to obtain total population. For 2025, annual population was not yet available in `BefolkningNy`; therefore, county population for 2025 was taken from SCB's monthly table `BefolkManadCKM` using the latest available month in 2025 at time of analysis and using **total age** and **both sexes combined**.

2.2 Statistical Analysis

For each pathogen, we fitted negative binomial regression models with **case counts** as the outcome, including **log-population as an offset** to model incidence rates, and adjusting for **region (fixed effects)**. To account for within-county temporal dependence (repeated observations of the same county over time), all standard errors, 95% confidence intervals, and p-values were computed using **cluster-robust (sandwich) standard errors** clustered at the county level (HC1 variant; `sandwich::vcovCL`). The COVID period (Pre-COVID, COVID, Post-COVID) was the primary exposure variable. Incidence rate ratios (IRRs) with 95% confidence intervals were calculated comparing:

1. COVID period vs. Pre-COVID (reference)
2. Post-COVID period vs. Pre-COVID (reference)
3. Post-COVID period vs. COVID (reference)

As a secondary analysis, we fitted an **interrupted time-series (ITS)** negative binomial model to annual county-level case counts (with log-population offset and region fixed effects), parameterizing the COVID-era disruptions as changes in **level** and **slope** at two interruption points (2020 and 2022). We use this model to visualize fitted trajectories and a counterfactual "no interruptions" trajectory representing continuation of the pre-2020 trend.

2.2.1 Model formulas

For the period-based regression, we modeled the expected case count μ_{ij} in region i and year j as:

$$\log(\mu_{ij}) = \log(P_{ij}) + \beta_0 + \beta_1 \cdot \mathbb{1}(\text{period}_j = \text{COVID}) + \beta_2 \cdot \mathbb{1}(\text{period}_j = \text{PostCOVID}) + \gamma_i$$

Equivalently, as a log-incidence model:

$$\log\left(\frac{\mu_{ij}}{P_{ij}}\right) = \beta_0 + \beta_1 \cdot \mathbb{1}(\text{period}_j = \text{COVID}) + \beta_2 \cdot \mathbb{1}(\text{period}_j = \text{PostCOVID}) + \gamma_i$$

Where:

- $Y_{ij} \sim \text{NegBin}(\mu_{ij}, \theta)$ = case count in region i and year j
- P_{ij} = population in region i and year j (offset term)
- γ_i = region fixed effect
- θ = dispersion parameter

Incidence rate ratios (IRRs) are obtained by exponentiating coefficients:

$$\text{IRR} = e^{\beta}$$

Statistical significance was assessed at $\alpha = 0.05$.

For the ITS model, we parameterized time as $t_j = \text{year}_j - 2015$, with step and slope terms at 2020 and 2022:

$$\text{covid_step}_j = \mathbb{1}(\text{year}_j \geq 2020), \quad \text{covid_slope}_j = \max(0, \text{year}_j - 2020)$$
$$\text{post_step}_j = \mathbb{1}(\text{year}_j \geq 2022), \quad \text{post_slope}_j = \max(0, \text{year}_j - 2022)$$

The ITS negative binomial model was:

$$\log(\mu_{ij}) = \log(P_{ij}) + \beta_0 + \beta_t t_j + \beta_{cs}\text{covid_step}_j + \beta_{csl}\text{covid_slope}_j + \beta_{ps}\text{post_step}_j + \beta_{psl}\text{post_slope}_j + \gamma_i$$

The "no interruptions" counterfactual shown in the figures corresponds to setting the interruption terms to zero (i.e., $\text{covid_step}_j = \text{covid_slope}_j = \text{post_step}_j = \text{post_slope}_j = 0$ for all j), yielding a continuation of the pre-2020 trend.

All analyses were performed in R version 4.5.1.

3 Results

3.1 Descriptive Statistics

3.1.1 Overall Case Counts by Period

Table 1: Summary of AMR-associated infection cases and incidence rates by pathogen and COVID period

pathogen	period	Total Cases	Mean Annual Cases	SD	Mean Incidence
ESBL/CARBA	Pre-COVID	703	140.6	35.7	1.39
ESBL/CARBA	COVID	265	132.5	6.4	1.27
ESBL/CARBA	Post-COVID	1411	352.8	93.8	3.34
MRSA	Pre-COVID	19704	3940.8	259.9	39.02
MRSA	COVID	5998	2999.0	159.8	28.80
MRSA	Post-COVID	14629	3657.2	272.2	34.62
VRE	Pre-COVID	1242	248.4	116.0	2.45
VRE	COVID	288	144.0	91.9	1.38
VRE	Post-COVID	1378	344.5	119.4	3.26

3.1.2 National Totals by Year

Table 2: National case totals and mean regional incidence rates by year

	ESBL/CARBA		VRE		MRSA	
Year	ESBL/CARBA Cases	ESBL/CARBA Inc.	VRE Cases	VRE Inc.	MRSA Cases	MRSA Inc.
2015	115	0.87	157	1.18	3879	38.53
2016	127	0.95	165	1.51	4392	52.62
2017	116	0.80	244	2.13	3721	39.45
2018	144	1.04	444	4.46	3856	39.74
2019	201	1.47	232	1.80	3856	38.24
2020	128	1.01	79	0.64	3112	29.40
2021	137	0.99	209	1.52	2886	28.61
2022	240	1.50	236	2.69	3331	30.55
2023	314	2.26	260	1.97	3545	32.24
2024	410	2.98	390	3.01	3936	32.32
2025	447	3.37	492	2.66	3817	32.63

3.2 Regression Analysis

3.2.1 Model Specification

Model equations and definitions are provided in the Methods section (Statistical Analysis).

3.2.2 Incidence Rate Ratios

We fitted log-linear regression models for each pathogen to estimate the change in incidence rates between periods, adjusting for regional variation.

Table 3: Incidence Rate Ratios (IRRs) comparing COVID periods, adjusted for region (cluster-robust SEs)

Pathogen	Comparison	IRR	95% CI	P-value
ESBL/CARBA	COVID vs Pre-COVID	0.95	(0.81 - 1.12)	0.568
ESBL/CARBA	Post-COVID vs COVID	2.54	(2.28 - 2.82)	<0.001
ESBL/CARBA	Post-COVID vs Pre-COVID	2.42	(2.10 - 2.79)	<0.001
MRSA	COVID vs Pre-COVID	0.70	(0.62 - 0.79)	<0.001
MRSA	Post-COVID vs COVID	1.13	(1.03 - 1.24)	0.013
MRSA	Post-COVID vs Pre-COVID	0.79	(0.69 - 0.90)	<0.001
VRE	COVID vs Pre-COVID	0.46	(0.25 - 0.85)	0.012
VRE	Post-COVID vs COVID	1.91	(1.12 - 3.25)	0.017
VRE	Post-COVID vs Pre-COVID	0.88	(0.52 - 1.50)	0.644

Note: IRR < 1 indicates lower incidence compared to reference period; IRR > 1 indicates higher incidence.

3.3 Interrupted time-series (ITS) model (secondary analysis)

To better align the model with the study question (a disruption during 2020-2021 and a potential rebound after 2022) and with temporal patterns, we additionally fitted an **interrupted time-series (ITS)** negative binomial model with:

- A baseline time trend (year),
- A COVID interruption at 2020 (step + slope change),
- A post-COVID interruption at 2022 (step + slope change),
- Region fixed effects and a population offset.

Table 4: Interrupted time-series effects (negative binomial + offset + region FE, cluster-robust SEs)

Pathogen	Effect	IRR	95% CI	P-value
ESBL/CARBA	COVID step (from 2020)	0.62	(0.49 - 0.79)	<0.001
ESBL/CARBA	COVID slope change (per year)	0.94	(0.74 - 1.19)	0.607
ESBL/CARBA	Post step (from 2022)	1.72	(1.34 - 2.22)	<0.001
ESBL/CARBA	Post slope change (per year)	1.15	(0.88 - 1.51)	0.307
VRE	COVID step (from 2020)	0.16	(0.08 - 0.31)	<0.001
VRE	COVID slope change (per year)	1.76	(0.73 - 4.24)	0.209
VRE	Post step (from 2022)	0.54	(0.11 - 2.65)	0.449
VRE	Post slope change (per year)	0.50	(0.20 - 1.21)	0.125
MRSA	COVID step (from 2020)	0.78	(0.71 - 0.85)	<0.001

MRSA	COVID slope change (per year)	1.00	(0.89 - 1.14)	0.953
MRSA	Post step (from 2022)	1.13	(0.94 - 1.35)	0.194
MRSA	Post slope change (per year)	1.05	(0.93 - 1.20)	0.422

Table 5: Model fit comparison by AIC (lower is better)

Pathogen	AIC (Period model)	AIC (ITS model)
ESBL/CARBA	1085.2	1020.8
VRE	1348.7	1344.5
MRSA	2231.8	2232.8

3.4 Sensitivity analysis: excluding 2025

Because the 2025 population denominator is derived from the latest available SCB monthly table (not a finalized annual total), we re-run both the period model and the ITS model on the 2015-2024 data only to assess the robustness of results.

3.4.1 Period model (2015-2024)

Table 6: Sensitivity: IRRs excluding 2025 (2015-2024 only), adjusted for region (cluster-robust SEs)

Pathogen	Comparison	IRR	95% CI	P-value
ESBL/CARBA	COVID vs Pre-COVID	0.94	(0.81 - 1.10)	0.460
ESBL/CARBA	Post-COVID vs COVID	2.30	(2.02 - 2.62)	<0.001
ESBL/CARBA	Post-COVID vs Pre-COVID	2.17	(1.91 - 2.46)	<0.001
MRSA	COVID vs Pre-COVID	0.70	(0.62 - 0.79)	<0.001
MRSA	Post-COVID vs COVID	1.12	(1.01 - 1.23)	0.027
MRSA	Post-COVID vs Pre-COVID	0.78	(0.68 - 0.90)	<0.001
VRE	COVID vs Pre-COVID	0.47	(0.26 - 0.84)	0.011
VRE	Post-COVID vs COVID	1.79	(1.03 - 3.12)	0.039
VRE	Post-COVID vs Pre-COVID	0.84	(0.48 - 1.49)	0.561

3.4.2 ITS model (2015-2024)

Table 7: Sensitivity: ITS effects excluding 2025 (2015-2024 only, cluster-robust SEs)

Pathogen	Effect	IRR	95% CI	P-value
ESBL/CARBA	COVID step (from 2020)	0.62	(0.49 - 0.79)	<0.001
ESBL/CARBA	COVID slope change (per year)	0.94	(0.74 - 1.19)	0.610

ESBL/CARBA	Post step (from 2022)	1.64	(1.29 - 2.08)	<0.001
ESBL/CARBA	Post slope change (per year)	1.23	(0.90 - 1.66)	0.192
VRE	COVID step (from 2020)	0.17	(0.09 - 0.31)	<0.001
VRE	COVID slope change (per year)	1.80	(0.76 - 4.24)	0.180
VRE	Post step (from 2022)	0.54	(0.11 - 2.53)	0.433
VRE	Post slope change (per year)	0.48	(0.19 - 1.23)	0.126
MRSA	COVID step (from 2020)	0.78	(0.71 - 0.85)	<0.001
MRSA	COVID slope change (per year)	1.00	(0.89 - 1.14)	0.958
MRSA	Post step (from 2022)	1.12	(0.94 - 1.33)	0.219
MRSA	Post slope change (per year)	1.07	(0.93 - 1.22)	0.350

Table 8: Sensitivity: AIC comparison excluding 2025 (lower is better)

Pathogen	AIC (Period model)	AIC (ITS model)
ESBL/CARBA	943.9	894.1
VRE	1220.7	1217.5
MRSA	2038.8	2040.3

3.5 Counterfactual trajectories (ITS)

The following figures show national observed incidence versus the fitted ITS trajectory, alongside a counterfactual trajectory. Although the ITS model is fitted to county-level counts (with region fixed effects and a population offset), the plotted trajectories are shown at the national level by aggregating observed and model-predicted county counts and converting to incidence using the corresponding national population.

- **No interruptions:** continuation of the pre-2020 trend.

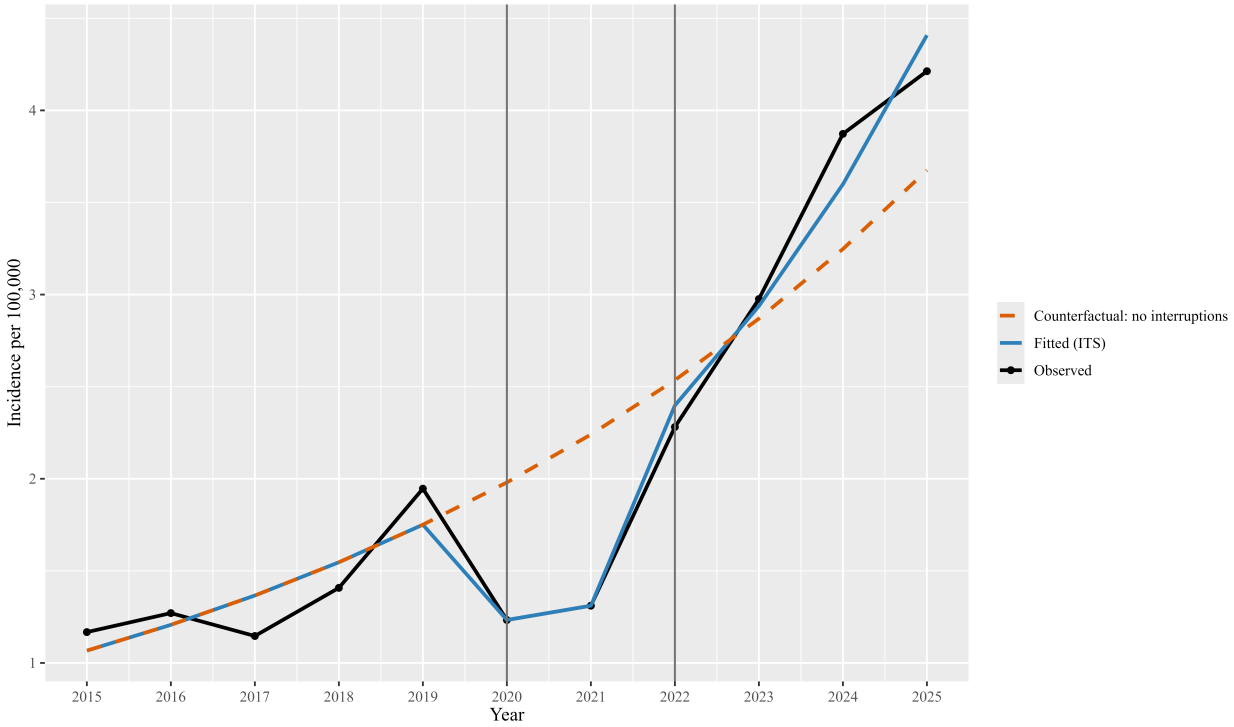

Figure 1: ESBL/CARBA: National observed vs fitted incidence trajectories with ITS counterfactual. Solid black: observed. Solid blue: fitted ITS. Dashed orange: counterfactual with no interruptions (continuation of pre-2020 trend). Vertical lines mark 2020 and 2022.

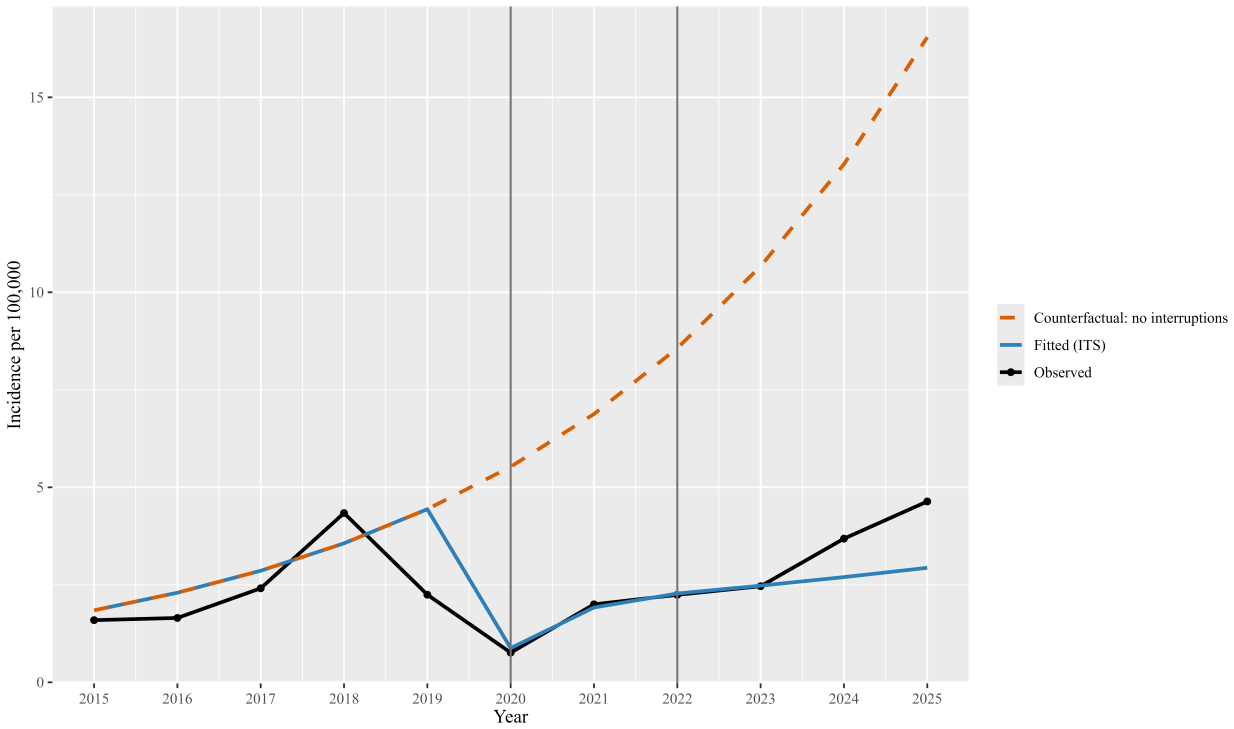

Figure 2: VRE: National observed vs fitted incidence trajectories with ITS counterfactual. Solid black: observed. Solid blue: fitted ITS. Dashed orange: counterfactual with no interruptions (continuation of pre-2020 trend). Vertical lines mark 2020 and 2022.

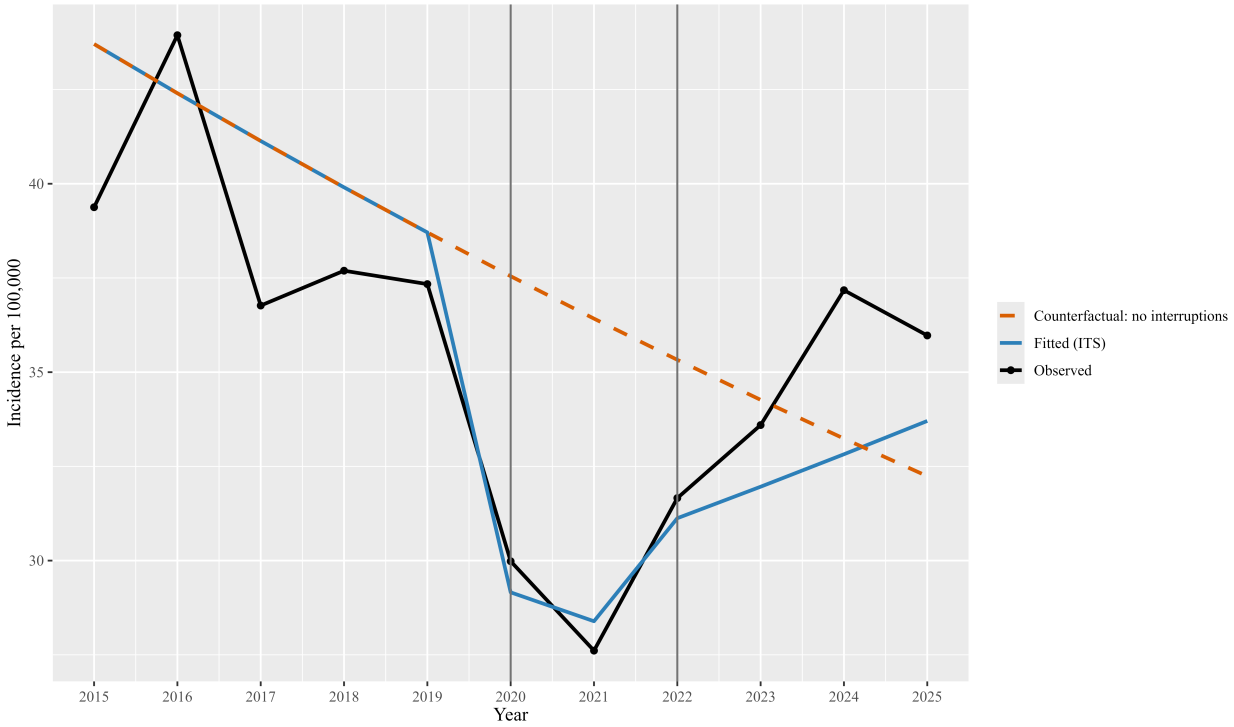

Figure 3: MRSA: National observed vs fitted incidence trajectories with ITS counterfactual. Solid black: observed. Solid blue: fitted ITS. Dashed orange: counterfactual with no interruptions (continuation of pre-2020 trend). Vertical lines mark 2020 and 2022.

3.5.1 Interpretation of Results

ESBL/CARBA (period model): Incidence during the COVID period was not significantly different from pre-COVID (IRR = 0.95 (0.81–1.12), p 0.568). However, the post-COVID period showed a substantial increase compared with both the COVID period (IRR = 2.54 (2.28–2.82), p <0.001; ~154% higher than COVID) and pre-COVID levels (IRR = 2.42 (2.10–2.79), p <0.001; ~142% higher than pre-COVID). This pattern suggests a delayed surge in ESBL/CARBA infections following the pandemic period. The ITS model indicates an immediate step decrease at 2020 followed by a significant step increase at 2022, consistent with the period model findings and visualized in the counterfactual trajectories.

VRE (period model): We observed a pronounced reduction during COVID versus pre-COVID (IRR = 0.46 (0.25–0.85), p 0.012). Post-COVID incidence rebounded strongly relative to the COVID period (IRR = 1.91 (1.12–3.25), p 0.017; ~91% higher than COVID), but was not significantly different from pre-COVID levels (IRR = 0.88 (0.52–1.50), p 0.644). This pattern suggests substantial disruption during the pandemic period followed by recovery to approximately pre-pandemic baseline. The ITS model shows a large immediate step decrease at 2020, consistent with the period model findings, and is visualized in the counterfactual trajectories.

MRSA (period model): COVID-period incidence was substantially lower than pre-COVID (IRR = 0.70 (0.62–0.79), p <0.001). The post-COVID period showed a modest rebound relative to the COVID period (IRR = 1.13 (1.03–1.24), p 0.013; ~13% higher than COVID), but incidence remained below pre-COVID levels (IRR = 0.79 (0.69–0.90), p <0.001; ~-21% lower than pre-COVID). This pattern suggests partial recovery but not a return to pre-pandemic baseline. The ITS model shows an immediate step decrease at 2020, but no significant step or slope changes at 2022, indicating a gradual recovery that remains below the counterfactual trajectory (see counterfactual plots).

3.6 Geographic Distribution

Figure 1 shows the regional distribution of mean incidence rates for each pathogen across the three time periods.

4 Discussion

This analysis reveals substantial disruptions to AMR-associated infection patterns in Sweden during and following the COVID-19 pandemic. The three pathogens showed varying patterns during 2020-2021, likely reflecting:

1. **Changes in healthcare utilization:** Reduced non-urgent healthcare visits may have affected detection and reporting of AMR infections
2. **Antimicrobial use patterns:** Pandemic-related changes in antibiotic prescribing and hospital infection control practices
3. **Travel restrictions:** Reduced international travel may have affected importation of resistant strains, particularly for ESBL/CARBA
4. **Healthcare-associated transmission:** Changes in hospital occupancy and infection prevention measures may have influenced MRSA transmission

The post-COVID recovery patterns differ between pathogens, suggesting pathogen-specific transmission dynamics and varying impacts of pandemic-related disruptions on different AMR-associated infections.

If surveillance artifacts were the sole explanation for the pandemic-era changes, broadly similar patterns would be expected across pathogens; the observed heterogeneity is consistent with pathogen-specific transmission dynamics in addition to surveillance and behavior changes.

4.1 Limitations

- Ecological study design precludes individual-level inference
- Potential surveillance artifacts during the pandemic period
- 2025 population denominators were derived from the latest available SCB monthly population table at the time of analysis (not a finalized annual total)
- Within-county temporal dependence is addressed via cluster-robust standard errors rather than explicit autocorrelation structures (e.g., AR terms); ITS level and slope terms partially capture structured time dependence, but residual autocorrelation may remain
- AMR surveillance may be influenced by testing practices and laboratory capacity, which may have varied during the pandemic

5 Conclusion

The COVID-19 pandemic significantly disrupted the epidemiology of AMR-associated infections in Sweden. While incidence rates showed varying patterns during the pandemic period, recovery patterns in 2022-2025 suggest differential impacts on transmission pathways and surveillance. Continued surveillance will be essential to determine whether pre-pandemic patterns will be fully restored and to monitor the long-term impact of pandemic-related changes on AMR epidemiology.

6 References

Swedish Public Health Agency. Disease Statistics (Sjukdomsstatistik). Available at: https://www.folkhalsomyndigheten.se/

7 Appendix: Regional Incidence Data

Table 9: Mean incidence rates (per 100,000) by region, pathogen, and period

	ESBL/CARBA			VRE			MRSA		
Region	Pre	COVID	Post	Pre	COVID	Post	Pre	COVID	Post
Blekinge	2.0	1.6	3.0	1.5	2.2	4.7	65.4	43.1	32.8
Dalarna	0.6	0.9	2.3	0.6	0.5	1.3	39.0	26.0	33.9
Gävleborg	0.7	1.7	3.8	1.7	2.6	19.6	30.9	29.4	30.7
Gotland	0.0	0.0	0.8	1.4	0.8	0.4	31.1	31.3	25.4
Halland	1.3	0.7	1.5	2.3	0.9	2.4	42.6	26.5	31.2
Jämtland	0.8	0.0	1.9	0.9	0.0	0.2	43.1	18.2	18.8
Jönköping	1.2	1.6	3.2	1.2	0.9	4.3	41.2	24.9	34.5
Kalmar	1.4	1.0	1.3	0.9	0.2	1.6	70.1	30.6	34.3
Kronoberg	0.9	1.2	2.8	1.4	0.5	1.0	54.3	36.7	44.4
Norrbotten	0.7	0.6	2.5	1.2	0.2	0.3	42.3	27.2	26.9
Orebro	0.9	1.1	2.9	6.1	1.0	1.1	46.1	44.4	29.9
Östergötland	0.6	0.5	1.5	0.3	5.0	1.7	31.9	25.7	29.9
Skåne	1.1	1.1	2.8	1.6	0.6	2.2	48.0	27.0	34.7
Södermanland	0.5	1.0	2.2	3.1	0.8	0.4	31.8	24.1	33.8
Stockholm	2.4	2.0	5.6	3.8	2.1	4.0	35.5	34.3	43.2
Uppsala	1.3	1.8	3.0	3.9	0.8	1.7	30.7	26.8	38.3
Värmland	0.8	0.7	2.0	2.3	0.3	0.4	47.1	20.5	24.2
Västerbotten	1.4	0.6	1.8	8.1	0.7	0.3	30.6	33.2	33.3
Västernorrland	0.8	0.6	1.4	0.2	0.6	0.2	40.2	19.8	25.3
Västmanland	0.6	1.1	3.4	2.3	0.4	1.3	43.7	36.5	34.7
Västra Götaland	1.4	1.1	3.3	1.7	1.4	5.0	30.4	22.7	30.2